# Lab Report: Image classification experiments

## 1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 32. All experiments were run on a single GPU with 24 GB of memory.

## 2 Results

Figure 1 shows the accuracy by dataset.

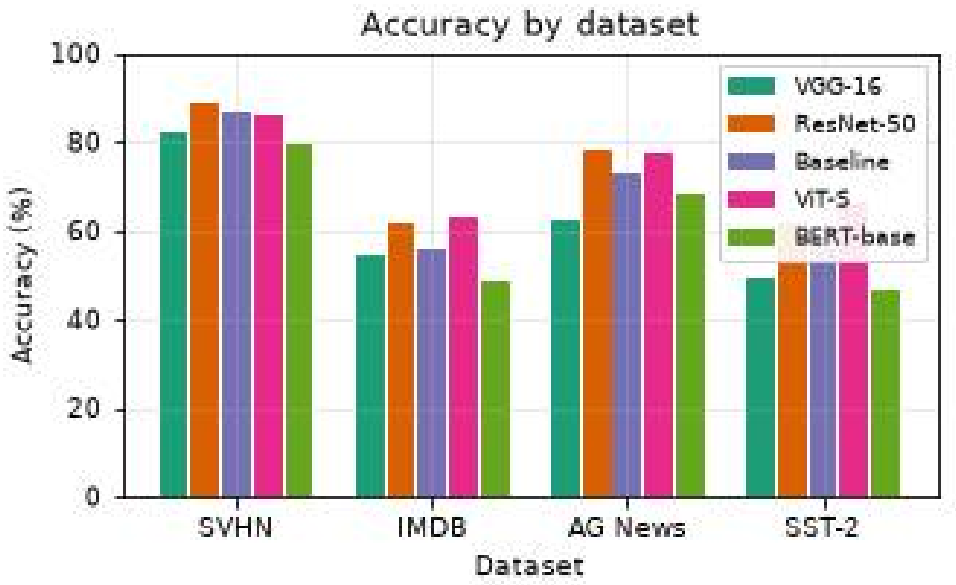



Figure 1: Accuracy by dataset for the models in this experiment.

ViT-S outperforms VGG-16 by 8.4 percentage points on IMDB (Figure 1). As shown in Figure 1, Baseline reaches about 60% accuracy on IMDB. As shown in Figure 1, ViT-S reaches 76.0% accuracy on SST-2.

Figure 2 shows the test accuracy during training.

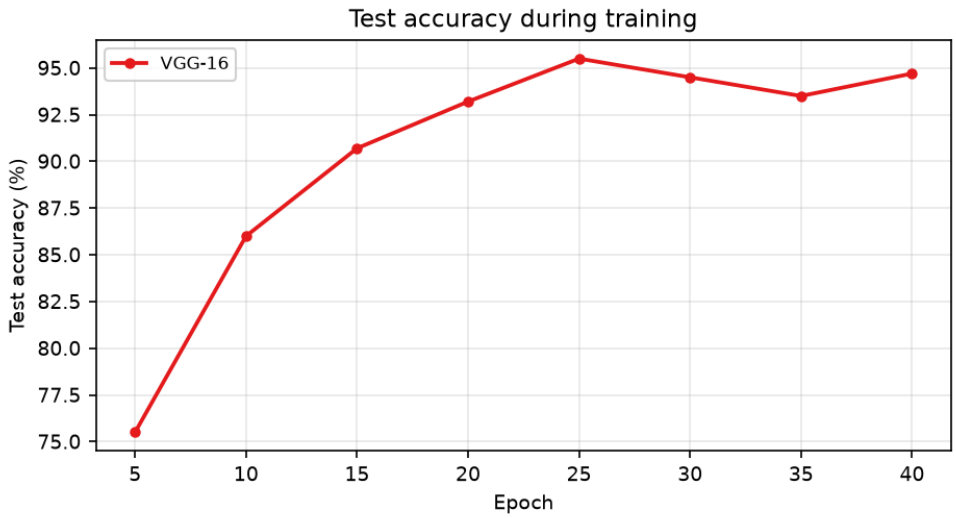



Figure 2: Test accuracy during training for the models in this experiment.

VGG-16 scores about 19 points higher on 40 than on 5 (Figure 2). At epoch 5, VGG-16 achieves about 80% test accuracy (Figure 2). At epoch 5, VGG-16 achieves 76.6% test accuracy. From epoch 10 to epoch 20, the test accuracy of VGG-16 improves by a relative 8.4% (Figure 2).

Figure 3 shows the f1 score by dataset.

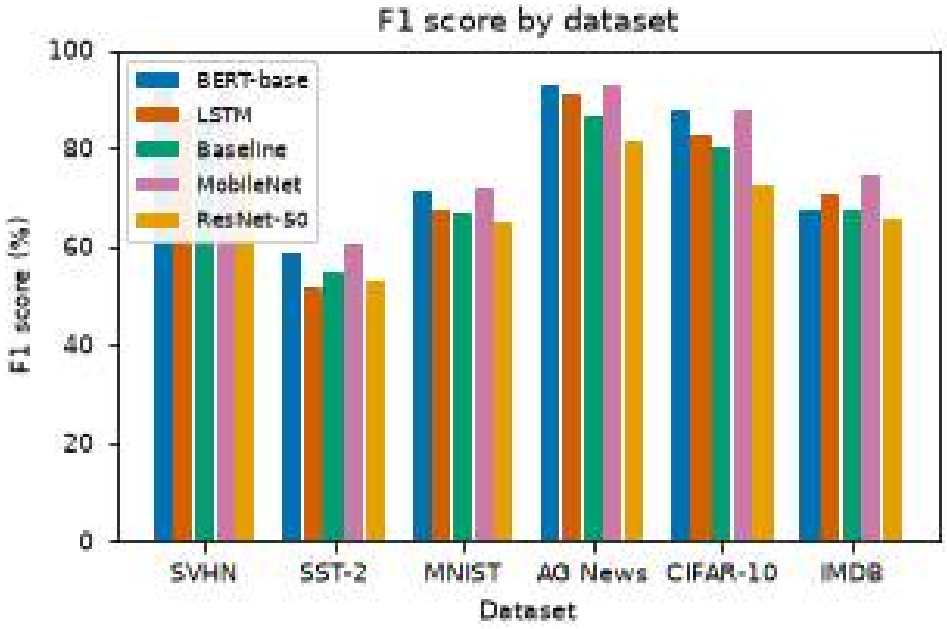



Figure 3: F1 score by dataset for the models in this experiment.

As shown in Figure 3, LSTM achieves 101.4% F1 score on SST-2. ResNet-50 achieves 75.0% F1 score on MNIST (Figure 3). ResNet-50 achieves 72.2% F1 score on CIFAR-10.

## 3 Discussion

On Fashion-MNIST, XGBoost obtains an accuracy of 74.3%. The differences between the models are discussed in the next section.